

Pankaj Yadav

Forward Deployed Software Engineer | Python, Linux & Computer Vision

Dubai, UAE | pankkaj64@gmail.com | +971 527314669 | [linkedin.com/in/pankkaj64](https://www.linkedin.com/in/pankkaj64) | github.com/Pankkaj64

PROFESSIONAL SUMMARY

Software engineer with **3+ years building and shipping production software** (including internship), mostly in **Python**. I take loosely defined requirements to working, deployed systems on my own: architecture, implementation, testing and deployment on **Linux**. I build connectors to external systems, debug live issues and write down what I learn. I also build real-time **computer vision and AI agents** on local models.

TECHNICAL SKILLS

Languages	Python (primary), TypeScript / JavaScript, SQL, Bash
Linux & Deployment	Scripted provisioning of fresh Debian / Ubuntu hosts, Docker & Compose, GitHub Actions / Gitea CI/CD, blue-green deploys with rollback, backup restore drills, load and soak tests
Systems & Integration	FastAPI, REST, WebSockets, Redis pub/sub and job queues, PostgreSQL, Alembic; connectors to Bitcoin Core JSON-RPC, LND gRPC / REST, Nostr relays, Telegram, WhatsApp and data APIs
Computer Vision & ML	YOLO detection, PaddleOCR, local vision-language models via Ollama (Gemma 3, Qwen3-VL), PyTorch, TensorFlow / Keras CNNs, Hugging Face fine-tuning (BERT, LoRA), scikit-learn
AI Agents	ReAct and tool-calling agents, RAG, LangChain, voice interfaces, Claude Code
Interfaces	React, Next.js, React Native / Expo, Tailwind CSS

WORK EXPERIENCE

Software Developer – Green Truth Technology LLC, Dubai	Aug 2025 – Present
• Own features end to end on a production Bitcoin payment platform that moves real funds (FastAPI, Next.js, Expo; 7-engineer team): I turn loose requirements into architecture, code, tests and deployment. • Built connectors to external systems and validated their data flows: Nostr relays with authentication, inbound event verification and ingest caps; Lightning wallet payments; media servers and BTC price oracles with automatic failover. • Deployed and ran the stack on Linux: scripted fresh-server provisioning, Docker Compose services, CI/CD pipelines, automated and verified database backups with a tested restore drill, and a load and soak test harness. • Made the system safe to run under failure: a durable Redis job queue with dead-letter records, fail-closed configuration that refuses unsafe settings at boot, purpose-bound signing keys and migration safety guards. • Debugged live issues and documented what I learned: wrote the incident, privacy-incident and deploy runbooks and 12 architecture decision records. • Shipped user-facing features fast and iterated on feedback: the community product (encrypted messaging with attachments and voice, group chat, stories) and the admin dashboard for burns, batches and transactions.	
Junior Software Engineer – Codensity Technology, Gurugram	Mar 2024 – May 2025
• Built and maintained full-stack applications for client projects with Python / Django and React: REST APIs, PostgreSQL schemas, production bug fixes and code reviews.	
Software Engineering Intern – Codensity Technology, Gurugram	Aug 2023 – Feb 2024

PROJECTS

Library Contents Claim Agent	Python, FastAPI, YOLO, PaddleOCR, Ollama VLMs, React
• A real-time voice-and-vision agent that inventories a room in one live camera sweep, with every model running locally on a laptop. A staged pipeline detects objects with YOLO, reads text with OCR, cross-checks with small vision-language models, and prices each item only from retrievable sources, with totals computed in code. Each sweep outputs a structured claim packet and an evidence bundle.	
Deep Learning & LLM Fine-tuning Lab	PyTorch, TensorFlow / Keras, Hugging Face, PEFT, Unsloth
• Built CNN image classifiers (71% vs 49% for a dense baseline), LSTM models and a fine-tuned BERT classifier. Benchmarked LoRA against full fine-tuning (91.1% vs 93.2% accuracy while training about 1% of the parameters) and explored quantization trade-offs.	
NostrAI – Autonomous Assistant	FastAPI, Flutter, Ollama, PostgreSQL, Redis, WebSockets
• A local-model agent with a ReAct tool loop and image understanding. Relay workers ingest live events from several relays into PostgreSQL and fan them out to clients over a WebSocket gateway on Redis pub/sub.	

EDUCATION

B.E. Computer Science – BMS Institute of Technology, Bengaluru	2019 – 2023
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CERTIFICATIONS

AI Engineering · Bitcoin Layer 1 · Lightning Network · Nostr Protocol — Green Truth Technology training